Supplementary Material of *Multistable active phases of complex contagions from temporal higher-order interactions*

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SUPPLEMENTARY NOTE 1: STABLE AND UNSTABLE FIXED POINTS

Fixed point condition through detailed balance

Imposing the stationary state condition in Eqs. (9)-(10) of the main text, i.e. that $ds_k/dt = 0$ and $df_{n,i}/dt = 0$, we obtain that

$$s_k = \frac{\mu}{\mu + mr}, \tag{S.1}$$

$$\begin{aligned}&(i+1)\big(\mu + \omega(1-I)\big)\, f_{n,i+1}\\ &= \left[i(\mu + \omega(1-I)) + (n-i)(\beta(n,i) + \rho + \omega I)\right] f_{n,i}\\ &-(n-i+1)(\beta(n,i-1) + \rho + \omega I) f_{n,i-1},\end{aligned} \tag{S.2}$$

where we assume the abuse of notation $s_k \equiv s_k^\star$ and $f_{n,i} \equiv f_{n,i}^\star$ for the shake of readability. We also know that in the stationary state $f_{n,i}$ must respect detailed balance:

$$(i+1)\big(\mu + \omega(1-I)\big)\, f_{n,i+1} \;=\; (n-i)\big(\beta(n,i) + \rho + \omega I\big)\, f_{n,i}. \tag{S.3}$$

Incorporating this condition in Eq. (S.2) leads to the iterative relation

$$f_{n,i} \;=\; f_{n,0} \frac{n!}{(n-i)!\, i!} \, \frac{\prod_{j=0}^{i-1} \left[\beta(n,j) + \rho + \omega I\right]}{(\mu + \omega(1-I))^i}, \tag{S.4}$$

where $f_{n,0} = 1 - \sum_{i=1}^{n} f_{n,i}$ due to the normalization constraint.

In the stationary state, the whole system can be rewritten in terms of r, ρ and I. Therefore, we can define a function $\mathcal{M}\left[\rho(r), I(r)\right]$, reading

$$\mathcal{M}\left[\rho(r), I(r)\right] = \frac{\sum_{n,i} \beta(n,i)(n-i) f_{n,i}(\rho, I) p_n}{\sum_{n,i} (n-i) f_{n,i}(\rho, I) p_n}, \tag{S.5}$$

where the dependency on ρ and I is in the $f_{n,i}(\rho, I)$, and that must fulfill the implicit relation $r = \mathcal{M}\left[\rho(r), I(r)\right]$. This relation can be used to solve numerically for the fixed points.

Jacobian of the dynamics

To evaluate the stability of the fixed points, we must also look at their Jacobian

$$J_r = \frac{dM}{dr} = \frac{\partial M}{\partial \rho}\frac{\partial \rho}{\partial r} + \frac{\partial M}{\partial I}\frac{\partial I}{\partial r}. \tag{S.6}$$

In the former expression, the derivatives $\frac{\partial \rho}{\partial r}$ and $\frac{\partial I}{\partial r}$ can be straightforwardly computed as

$$\frac{\partial \rho}{\partial r} \;=\; \frac{u_2}{v_2} + r\left[\frac{1}{v_2}\frac{du_2}{dr} - \frac{u_2}{v_2^2}\frac{dv_2}{dr}\right], \tag{S.7}$$

$$\frac{\partial I}{\partial r} \;=\; -\sum_k k \frac{ds_k}{dr} g_k, \tag{S.8}$$

where

$$\begin{aligned}
u_2 \;&=\; \sum_k k(k-1) s_k(r) g_k,\\
v_2 \;&=\; \sum_k k s_k(r) g_k,\\
\frac{du_2}{dr} \;&=\; \sum_k k(k-1)\frac{ds_k}{dr} g_k,\\
\frac{dv_2}{dr} \;&=\; \sum_k k\frac{ds_k}{dr} g_k,\\
\frac{ds_k}{dr} \;&=\; -\frac{\mu k}{(\mu + kr)^2}.
\end{aligned}$$

However, the derivatives $\frac{\partial M}{\partial \rho}$ and $\frac{\partial M}{\partial I}$ are more difficult to be computed as they explicitly depend on $\frac{df_{n,i}}{d\rho}$ and $\frac{df_{n,i}}{dI}$ as

$$\frac{\partial M}{\partial \rho} = \frac{1}{v_1}\frac{du_1}{d\rho} - \frac{u_1}{v_1^2}\frac{dv_1}{d\rho}, \tag{S.9}$$

$$\frac{\partial M}{\partial I} = \frac{1}{v_1}\frac{du_1}{dI} - \frac{u_1}{v_1^2}\frac{dv_1}{dI}, \tag{S.10}$$

$$\tag{S.11}$$

where

$$\begin{aligned}
u_1 &= \sum_{n,i} \beta(n,i)(n-i) f_{n,i}(\rho, I) p_n, \\
v_1 &= \sum_{n,i} (n-i) f_{n,i}(\rho, I) p_n, \\
\frac{du_1}{d\rho} &= \sum_{n,i} \beta(n,i)(n-i) \frac{df_{n,i}}{d\rho} p_n, \\
\frac{du_1}{dI} &= \sum_{n,i} \beta(n,i)(n-i) \frac{df_{n,i}}{dI} p_n, \\
\frac{dv_1}{d\rho} &= \sum_{n,i} (n-i) \frac{df_{n,i}}{d\rho} p_n, \\
\frac{dv_1}{dI} &= \sum_{n,i} (n-i) \frac{df_{n,i}}{dI} p_n.
\end{aligned}$$

In order to compute the derivatives $\frac{df_{n,i}}{d\rho}$ and $\frac{df_{n,i}}{dI}$, we rewrite the stationary recursive relation in terms of a closed form and its recursive factor. From Eq. (S.4) we first define $B_j = \beta(n,j) + \rho + \omega I$ and $A = \mu + \omega(1-I)$, yielding

$$f_{n,i} = f_{n,0}\binom{n}{i}\prod_{j=0}^{i-1}\frac{B_j}{A} \quad \Leftrightarrow \quad f_{n,i} = \frac{g_{n,i}}{Z_n}, \tag{S.12}$$

with

$$g_{n,i} = \binom{n}{i}\prod_{j=0}^{i-1}\frac{B_j}{A}, \qquad Z = \sum_{l=0}^{k} g_{n,l}, \tag{S.13}$$

where we have set as normalization $g_{n,0} = 1$ (since $f_{n,0}$ is defined by normalization). Therefore, we can express the desired derivatives in terms of $\theta = \{\rho, I\}$ as

$$\frac{df_{n,i}}{d\theta} = \frac{g'_{n,i}Z_n - g_{n,i}Z'_n}{Z_n^2} = f_{n,i}\left(\frac{g'_{n,i}}{g_{n,i}} - \frac{Z'_n}{Z_n}\right) = f_{n,i}\left(\frac{g'_{n,i}}{g_{n,i}} - \sum_l f_{n,l}\frac{g'_{n,l}}{g_{n,l}}\right) = f_{n,i}\left(\Delta^{\theta}_{n,i} - \sum_l f_{n,l}\Delta^{\theta}_{n,l}\right), \tag{S.14}$$

where we have used that $\frac{Z'_n}{Z_n} = \frac{\sum_l g'_{n,l}}{\sum_l g_{n,l}} = \sum_l \frac{g_{n,l}}{Z_n}\frac{g'_{n,l}}{g_{n,l}} = \sum_l f_{n,l}\frac{g'_{n,l}}{g_{n,l}}$, and we have defined

$$\Delta^{\theta}_{n,i} \equiv \frac{g'}{g} = \frac{1}{g}\frac{dg}{d\theta} = \frac{d}{d\theta}\ln g_{n,i}. \tag{S.15}$$

Expressing the former quantity in terms of the derivative of the logarithm eases the mathematical derivation, as it allows for splitting the terms of Eq. (S.13) as follows:

$$\ln(g_{n,i}) = \ln\binom{n}{i} + \sum_{j=0}^{i-1}\ln B_j - i\ln A, \tag{S.16}$$

$$\frac{d}{d\theta}\ln(g_{n,i}) = \sum_{j=0}^{i-1}\frac{1}{B_j}\frac{dB_j}{d\theta} - i\frac{1}{A}\frac{dA}{d\theta}, \tag{S.17}$$

being $\frac{dB_j}{d\rho} = 1$, $\frac{dB_j}{dI} = \omega$, $\frac{dA}{d\rho} = 0$ and $\frac{dA}{dI} = -\omega$. Therefore, we reach both closed expressions for $\frac{df_{n,i}}{d\theta}$, $\theta = \{\rho, I\}$ as

$$\frac{df_{n,i}}{d\theta} = f_{n,i}\left(\Delta^{\theta}_{n,i} - \sum_{l} f_{n,l}\Delta^{\theta}_{n,l}\right) \quad \text{with} \quad \theta = \{\rho, I\}, \tag{S.18}$$

$$\Delta^{\rho}_{n,i} = \sum_{j=0}^{i-1} \frac{1}{\beta(n,j) + \rho + \omega I}, \tag{S.19}$$

$$\Delta^{I}_{n,i} = \omega\Delta^{\rho}_{n,i} + \frac{\omega i}{\mu + \omega(1-I)}. \tag{S.20}$$

The expressions above provide all the required ingredients to evaluate the Jacobian J_r at any stationary solution of the AME system. The sign of J_r fully determines the linear stability of the corresponding fixed point: $J_r < 0$ identifies a stable branch, $J_r > 0$ an unstable one, and the condition $J_r = 0$ detects the onset of saddle–node bifurcations.

SUPPLEMENTARY NOTE 2: CONDITIONS FOR THE INVASION THRESHOLD AND THE TRICRITICAL POINT

The invasion threshold is the critical point that marks the limit of the domain of validity of the solution $r = \mathcal{M}\left[\rho(r), I(r)\right]$. It arises when r becomes to $\mathcal{M}\left[\rho(r), I(r)\right]$, i.e. when

$$\frac{d\mathcal{M}}{dr} = \frac{\partial M}{\partial \rho}\frac{\partial \rho}{\partial r} + \frac{\partial M}{\partial I}\frac{\partial I}{\partial r} = 1. \tag{S.21}$$

Moreover, to have tricritical points (i.e., where absorbing, active, and bistable phases merge), the nontrivial solutions must be degenerated, which requires that the second derivative is null.

$$\frac{d^2M}{dr^2} = \frac{\partial^2 M}{\partial \rho^2}\left(\frac{d\rho}{dr}\right)^2 + 2\frac{\partial^2 M}{\partial\rho\partial I}\frac{d\rho}{dr}\frac{dI}{dr} + \frac{\partial^2 M}{\partial I^2}\left(\frac{dI}{dr}\right)^2 + \frac{\partial M}{\partial \rho}\frac{d^2\rho}{dr^2} + \frac{\partial M}{\partial I}\frac{d^2 I}{dr^2} = 0. \tag{S.22}$$

Derivatives

In the limit of $r \to 0$, it is possible to derive a semi-analytical expression for both the invasion threshold and the tricritical point. In this limit, $s_k \to 1$, $\rho \to 0$, $I \to 0$ and $f_{n,i} \to \delta_{i,0}$, i.e. all nodes are susceptible. Some of the derivatives are simple, as

$$\left.\frac{d\rho}{dr}\right|_{(0,0)} = \frac{\langle k(k-1)\rangle}{\langle k\rangle}, \tag{S.23}$$

$$\left.\frac{dI}{dr}\right|_{(0,0)} = \frac{\langle k\rangle}{\mu}, \tag{S.24}$$

$$\left.\frac{d^2\rho}{dr^2}\right|_{(0,0)} = \frac{2}{\mu}\left(\frac{\langle k^2\rangle^2}{\langle k\rangle^2} - \frac{\langle k^3\rangle}{\langle k\rangle}\right), \tag{S.25}$$

$$\left.\frac{d^2 I}{dr^2}\right|_{(0,0)} = -2\frac{\langle k^2\rangle}{\mu^2}, \tag{S.26}$$

$$\tag{S.27}$$

However, to solve the core derivatives $\frac{\partial M}{\partial \rho}|_{r\to 0}$ and $\frac{\partial M}{\partial I}|_{r\to 0}$ we realize that the denominator of

$$\mathcal{M}\left[\rho(r), I(r)\right] = \frac{\sum_{n,i}\beta(n,i)(n-i)f_{n,i}(\rho,I)p_n}{\sum_{n,i}(n-i)f_{n,i}(\rho,I)p_n}, \tag{S.28}$$

becomes $\langle n\rangle$, and in the numerator all terms are small, since $\beta(n,0) = 0$, which is the one that corresponds to $\delta_{i,0}$. Therefore, we approximate that

$$\frac{d\mathcal{M}}{d\theta}|_{r\to 0} = \frac{1}{\langle n\rangle}\sum_{n,i}\beta(n,i)(n-i)p_n h^{\theta}_{n,i}, \quad \text{with } h^{\theta}_{n,i} = \frac{f_{n,i}}{d\theta}|_{r\to 0}, \quad \theta = \{\rho, I\}. \tag{S.29}$$

For the second derivative we have that

$$\frac{d^2\mathcal{M}}{d\theta^2} = \frac{(u''v + u'v' - u'v' - uv'')v^2 - (u'v - uv')2vv'}{v^4} \tag{S.30}$$

$$= \frac{v\left(u''v - uv''\right) + 2v'\left(uv' - u'v\right)}{v^3}. \tag{S.31}$$

Now, in the limit $\rho \to 0$ we have $u(0) = 0$ and $v(0) = \langle n\rangle$. Substituting these values, all terms proportional to u vanish, leaving

$$\left.\frac{d^2\mathcal{M}}{d\theta^2}\right|_{(0,0)} = \frac{u''v^2 - 2vu'v'}{v^3} \tag{S.32}$$

$$= \frac{1}{\langle n\rangle}\frac{d^2u}{d\theta^2} - \frac{2}{\langle n\rangle^2}\frac{du}{d\theta}\frac{dv}{d\theta}, \tag{S.33}$$

where, after defining $h^{\theta}_{n,i} = \frac{df_{n,i}}{d\theta}$ and $l^{\theta}_{n,i} = \frac{d^2 f_{n,i}}{d\theta^2}$:

$$\begin{aligned}\frac{du}{d\theta} &= \sum_{n,i} \beta(n,i)(n-i)p_n h^{\theta}_{n,i},\\ \frac{dv}{d\theta} &= \sum_{n,i} (n-i)p_n h^{\theta}_{n,i},\\ \frac{d^2u}{d\theta^2} &= \sum_{n,i} \beta(n,i)(n-i)p_n l^{\theta}_{n,i}.\end{aligned} \tag{S.34}$$

Moreover, the derivative $\left.\frac{d^2\mathcal{M}}{d\rho dI}\right|_{(0,0)}$ requires special treatment, as

$$\left.\frac{d^2\mathcal{M}}{d\rho dI}\right|_{(0,0)} = \frac{1}{\langle n\rangle}\frac{d^2u}{d\rho d\theta} - \frac{1}{\langle n\rangle^2}\left(\frac{du}{d\rho}\frac{dv}{dI} + \frac{du}{dI}\frac{dv}{d\rho}\right), \tag{S.35}$$

and therefore we define $l^{\mathsf{mix}}_{n,i} = \frac{d^2 f_{n,i}}{d\rho dI}$ and

$$\frac{d^2u}{d\rho dI} = \sum_{n,i} \beta(n,i)(n-i)p_n l^{\mathsf{mix}}_{n,i}. \tag{S.36}$$

Overall, we need to compute $h^{\rho}_{n,i}, h^{I}_{n,i}, l^{\rho}_{n,i}, l^{I}_{n,i}, l^{\mathsf{mix}}_{n,i}$. In order to do so we start from the stationary relation between the $f_{n,i}$'s reported in Eq. (S.2), knowing also that in the stationary state $f_{n,i}$ must respect detailed balance condition in Eq. (S.3). Moreover, since probabilities must satisfy $\sum_i f_{n,i} = 1$ for each n, the derivatives fulfill

$$\sum_i h^{\theta}_{n,i} = 0, \qquad \sum_i l^{\theta}_{n,i} = 0, \tag{S.37}$$

so that the base values can be written as

$$h^{\theta}_{n,0} = -\sum_{i\geq 1} h^{\theta}_{n,i}, \qquad l^{\theta}_{n,0} = -\sum_{i\geq 1} l^{\theta}_{n,i}, \qquad \theta \in \{\rho, I, \mathsf{mix}\}. \tag{S.38}$$

First derivative in terms of ρ: $h^{\rho}_{n,i}$. The derivative reads

$$\begin{aligned}&(i+1)\big(\mu + \omega(1-I)\big)\, h^{\rho}_{n,i+1}\\ &= (n-i)f_{n,i} + \left[i(\mu + w(1-I)) + (n-i)(\beta(n,i) + \rho + \omega I)\right] h^{\rho}_{n,i}\\ &-(n-i+1)f_{n,i-1} - (n-i+1)(\beta(n,i-1) + \rho + \omega I)h^{\rho}_{n,i-1},\end{aligned} \tag{S.39}$$

and after the detailed balance and substituting that $\rho = I = f_n = 0$ we obtain that:

- if $i = 0$: $h^{\rho}_{n,1} = \frac{n}{\mu+\omega}$
- if $i > 0$: $h^{\rho}_{n,i+1} = \frac{(n-i)}{(i+1)}\frac{\beta(n,i)}{(\mu+\omega)}h^{\rho}_{n,i}$

First derivative in terms of I: $h^{I}_{n,i}$. The derivative reads

$$\begin{aligned}&(i+1)(-\omega)f_{n,i+1} + (i+1)\big(\mu + \omega(1-I)\big)\, h^{I}_{n,i+1}\\ &= (-i\omega + (n-i)\omega)f_{n,i} + \left[i(\mu + \omega(1-I)) + (n-i)(\beta(n,i) + \rho + \omega I)\right] h^{I}_{n,i}\\ &-(n-i+1)\omega f_{n,i-1} - (n-i+1)(\beta(n,i-1) + \rho + \omega I)h^{I}_{n,i-1},\end{aligned} \tag{S.40}$$

and after the detailed balance and substituting that $\rho = I = f_n = 0$ we obtain that:

- if $i=0$: $h^I_{n,1} = \frac{n\omega}{\mu+\omega}$
- if $i>0$: $h^I_{n,i+1} = \frac{(n-i)}{(i+1)}\frac{\beta(n,i)}{(\mu+\omega)}h^I_{n,i}$

Second derivative in terms of ρ: $l^\rho_{n,i}$. The derivative reads

$$\begin{aligned}&(i+1)\big(\mu+\omega(1-I)\big)\,l^\rho_{n,i+1}\\&=2(n-i)h^\rho_{n,i}-2(n-i+1)h^\rho_{n,i-1}+\left[i(\mu+\omega(1-I))+(n-i)(\beta(n,i)+\rho+\omega I)\right]l^\rho_{n,i}\\&-(n-i+1)(\beta(n,i-1)+\rho+\omega I)l^\rho_{n,i-1},\end{aligned}\tag{S.41}$$

and substituting that $\rho=I=f_n=0$ we obtain that:

- if $i=0$: $l^\rho_{n,1} = \frac{2n}{\mu+\omega}h^\rho_{n,0}$
- if $i>0$:

$$\begin{aligned}(i+1)\big(\mu+\omega\big)\,l^\rho_{n,i+1} \;=\;& 2(n-i)h^\rho_{n,i}-2(n-i+1)h^\rho_{n,i-1}\\&+\left[i(\mu+\omega)+(n-i)\beta(n,i)\right]l^\rho_{n,i}\\&-(n-i+1)\beta(n,i-1)l^\rho_{n,i-1},\end{aligned}$$

Second derivative in terms of I: $l^I_{n,i}$. The derivative reads

$$\begin{aligned}&-2\omega(i+1)h^I_{n,i+1}+(i+1)\big(\mu+\omega(1-I)\big)\,l^I_{n,i+1}\\&=2(-i\omega+(n-i)\omega)h^I_{n,i}-2(n-i+1)h^I_{n,i-1}+\left[i(\mu+\omega(1-I))+(n-i)(\beta(n,i)+\rho+\omega I)\right]l^I_{n,i}\\&-2(n-i+1)\omega h^I_{n,i-1}-(n-i+1)(\beta(n,i-1)+\rho+\omega I)l^I_{n,i-1},\end{aligned}\tag{S.42}$$

and substituting that $\rho=I=f_n=0$ we obtain that:

- if $i=0$: $l^I_{n,1} = \frac{2\omega}{\mu+\omega}\left(nh^I_{n,0}+h^I_{n,1}\right)$
- if $i>0$:

$$\begin{aligned}(i+1)\big(\mu+\omega\big)\,l^I_{n,i+1} \;=\;& 2\omega\left[-i+(n-i)\right]h^I_{n,i}-2\omega(n-i+1)h^I_{n,i-1}\\&+\left[i(\mu+\omega)+(n-i)\beta(n,i)\right]l^I_{n,i}\\&-(n-i+1)\beta(n,i-1)l^I_{n,i-1}\\&+2\omega(i+1)h^I_{n,i+1},\end{aligned}$$

Second derivative in terms of I and ρ (mixed derivative): $l^{\mathsf{mix}}_{n,i}$. The derivative reads

$$\begin{aligned}&-\omega(i+1)h^\rho_{n,i+1}+(i+1)\big(\mu+\omega(1-I)\big)\,l^{\mathsf{mix}}_{n,i+1}\\&=(n-i)h^I_{n,i}+\omega\left[-i+(n-i)\right]h^\rho_{n,i}+\left[i(\mu+\omega(1-I))+(n-i)(\beta(n,i)+\rho+\omega I)\right]l^{\mathsf{mix}}_{n,i}\\&-(n-i+1)\left[h^I_{n,i-1}+\omega h^\rho_{n,i-1}\right]-(n-i+1)(\beta(n,i-1)+\rho+\omega I)l^{\mathsf{mix}}_{n,i-1},\end{aligned}\tag{S.43}$$

and substituting that $\rho=I=f_n=0$ we obtain that:

- if $i=0$: $l^{\mathsf{mix}}_{n,1} = \frac{1}{\mu+\omega}\left(nh^I_{n,0}+n\omega h^\rho_{n,0}+\omega h^\rho_{n,1}\right)$
- if $i>0$:

$$\begin{aligned}(i+1)\big(\mu+\omega\big)\,l^{\mathsf{mix}}_{n,i+1} \;=\;& (n-i)h^I_{n,i}+\omega\left[-i+(n-i)\right]h^\rho_{n,i}-(n-i+1)\left[h^I_{n,i-1}+wh^\rho_{n,i-1}\right]\\&+\left[i(\mu+\omega)+(n-i)\beta(n,i)\right]l^{\mathsf{mix}}_{n,i}\\&-(n-i+1)\beta(n,i-1)l^{\mathsf{mix}}_{n,i-1}\\&+\omega(i+1)h^\rho_{n,i+1},\end{aligned}$$

General expressions

Incorporating the expressions of the derivatives into Eq. (S.21), we obtain the following closed expression for the invasion threshold:

$$\left\langle \sum_{i=1}^{n} \frac{n!}{(n-i-1)!\,i!} \, (\mu+\omega)^{-i} \prod_{j=1}^{i} \beta(n,j) \right\rangle \left(\frac{\langle k(k-1)\rangle}{\langle k\rangle\langle n\rangle} + \frac{\omega}{\mu}\frac{\langle k\rangle}{\langle n\rangle} \right) = 1. \tag{S.44}$$

Similarly, incorporating the expressions of the derivatives into Eq. (S.22), we obtain the following closed expression for the tricritical point.

$$F\frac{\langle k(k-1)\rangle^2}{\langle k\rangle^2} + 2G\frac{\langle k(k-1)\rangle}{\langle k\rangle}\frac{\langle k\rangle}{\mu} + H\frac{\langle k\rangle^2}{\mu^2} + J\left[\frac{2}{\mu}\left(\frac{\langle k^2\rangle^2}{\langle k\rangle^2} - \frac{\langle k^3\rangle}{\langle k\rangle}\right) - 2\omega\frac{\langle k^2\rangle}{\mu^2}\right] = 0. \tag{S.45}$$

where $F \equiv F\left[p_n, \beta(n,i), \omega\right] = \left.\frac{\partial^2 M}{\partial \rho^2}\right|_{(\rho,I)\to(0,0)}$, $G \equiv G\left[p_n, \beta(n,i), \omega\right] = \left.\frac{\partial^2 M}{\partial \rho \partial I}\right|_{(\rho,I)\to(0,0)}$, $H \equiv H\left[p_n, \beta(n,i), \omega\right] = \left.\frac{\partial^2 M}{\partial I^2}\right|_{(\rho,I)\to(0,0)}$, $J \equiv J\left[p_n, \beta(n,i), \omega\right] = \left.\frac{\partial M}{\partial \rho}\right|_{(\rho,I)\to(0,0)}$. Moreover, recalling from the invasion threshold that $\left.\frac{\partial M}{\partial \rho}\right|_{(\rho,I)\to(0,0)} = \left(\frac{\partial \rho}{\partial r} + \omega\frac{\partial I}{\partial r}\right)^{-1}$, we can rewrite the expression as

$$0 \;=\; F\frac{\langle k(k-1)\rangle^2}{\langle k\rangle^2} + 2G\frac{\langle k(k-1)\rangle}{\langle k\rangle}\frac{\langle k\rangle}{\mu} + H\frac{\langle k\rangle^2}{\mu^2} + 2\frac{\frac{1}{\mu}\Big(\langle k^2\rangle^2 - \langle k^3\rangle\langle k\rangle\Big) - \frac{\omega}{\mu^2}\langle k^2\rangle\langle k\rangle^2}{\langle k(k-1)\rangle\langle k\rangle + \frac{\omega}{\mu}\langle k\rangle^3}. \tag{S.46}$$

SUPPLEMENTARY NOTE 3: INVASION THRESHOLD

Limiting cases of the invasion threshold

We can obtain an explicit expression of the invasion threshold by imposing that $i = 1$ in Eq. (S.44), i.e. assuming that at the onset of the epidemic all infectious groups have only one infectious individual. The obtained expression reads

$$\lambda_c \approx \frac{\mu+\omega}{\langle n(n-1)\rangle}\left[\frac{\langle k(k-1)\rangle}{\langle k\rangle\langle n\rangle}+\frac{\omega}{\mu}\frac{\langle k\rangle}{\langle n\rangle}\right]^{-1}. \tag{S.47}$$

Quenched limit

We now derive the quenched limit reported in Eq. (1) of the main text for the case in which each individual participates in a single group at a time ($g(k) = \delta_{k,1}$). In this case $\langle k\rangle = 1$ and $\langle k(k-1)\rangle = 0$. Substituting these relations into Eq. (S.47), the expression inside the brackets simplifies to $\omega/(\mu\,\langle n\rangle)$. Equation (S.47) thus becomes

$$\lambda_c \approx \frac{\mu+\omega}{\langle n(n-1)\rangle}\cdot\frac{\mu\,\langle n\rangle}{\omega}. \tag{S.48}$$

In the limit $\omega \to 0$, the numerator satisfies $\mu + \omega \sim \mu$, so that

$$\lambda_c \sim \frac{\mu^2\,\langle n\rangle}{\omega\,\langle n(n-1)\rangle}, \qquad \text{as } \omega \to 0. \tag{S.49}$$

Therefore, the invasion threshold diverges in the quenched limit,

$$\lim_{\omega\to 0}\lambda_c(\omega) = \infty, \tag{S.50}$$

showing that infinitesimal adoption cannot invade when the higher-order structure is static and fragmented (each individual belongs to a single group only).

Annealed limit

To compute the annealed (mean-field) limit, we identify the dominant terms in Eq. (S.47) as $\omega \to \infty$. In the numerator, $\mu + \omega \sim \omega$, while the term inside the brackets becomes $\omega\langle k\rangle/(\mu\langle n\rangle)$. Substituting these asymptotic forms into Eq. (S.47) yields to the invasion threshold in the annealed limit

$$\lim_{\omega\to\infty}\lambda_c(\omega) = \frac{\mu\,\langle n\rangle}{\langle k\rangle\,\langle n(n-1)\rangle}, \tag{S.51}$$

that corresponds to the mean-field limit of the dynamics.

Closed expression of the invasion threshold

We start from the invasion threshold condition in Eq. (S.44), and consider the case of fixed group size ($p(n') = \delta_{n',n}$), so that $\langle n\rangle = n$ and the angular brackets over n become trivial. Defining $x \equiv \lambda_c/(\mu+\omega)$, and using $\prod_{j=1}^{i} j^\nu = (i!)^\nu$ and $\frac{n!}{(n-i-1)!\,i!} = n\binom{n-1}{i}$ for $i = 1, ..., n-1$, we can define the polynomial

$$F_n(x) = n\sum_{i=1}^{n-1}\binom{n-1}{i}(i!)^\nu\,x^i, \tag{S.52}$$

so that Eq. (S.44) becomes

$$F_n(x) = \left[\frac{\langle k(k-1)\rangle}{\langle k\rangle\,n}+\frac{\omega}{\mu}\frac{\langle k\rangle}{n}\right]^{-1}, \tag{S.53}$$

which is a $(n-1)$-degree polynomial equation for x which implicity defines the invasion threshold via $\lambda_c = x(\mu+\omega)$. Note that starting from Eq. (S.53) we can recover the quenched and annealed limits.

For fixed group size $p(n) = \delta_{n,3}$ and fixed membership $g(\kappa) = \delta_{\kappa,k}$, the threshold condition in Eq. (S.53) reduces to

$$x\big(2^\nu x + 2\big) = \frac{\mu}{k\omega + \mu(k-1)}. \tag{S.54}$$

Solving this quadratic equation for x keeping the physical (positive) root and recovering that $\lambda_c = x(\mu+\omega)$, we obtain the critical line

$$\lambda_c(\omega; k) = \frac{\mu+\omega}{2^\nu}\left[\sqrt{1 + \frac{2^\nu \mu}{k\omega + \mu(k-1)}} - 1\right]. \tag{S.55}$$

Optimal plasticity for maximizing adoption

To locate the minimum of the critical line we differentiate Eq. (S.55) with respect to ω and set the derivative to zero. To do so, we can define $D(\omega) = (k-1) + \frac{k\omega}{\mu}$ and $Y(\omega) = \sqrt{1 + \frac{2^\nu}{D(\omega)}}$. Differentiating Eq. (S.55) gives

$$\frac{d\lambda_c}{d\omega} = \frac{1}{2^\nu}\left[Y - 1 - \frac{(\mu+\omega)k}{2\mu\, D(\omega)^2\, Y}\right].$$

The minimum of $\lambda_c(\omega; k)$ satisfies $d\lambda_c/d\omega = 0$, which leads to $2\mu D(\omega)\, Y(\omega) = k(\mu+\omega)\big[Y(\omega)+1\big]$. Finally, using the identity $Y^2 = 1 + \frac{2^\nu}{D}$, and substituting $D(\omega) = (k-1) + k\omega/\mu$, we obtain an algebraic equation whose unique positive solution yields the optimal mixing rate,

$$\omega^\star(k) = \mu\, \frac{2^{\nu/2+1} + 2^{\nu+1} - 4 - k\,(2^\nu - 4)}{k\,(2^\nu - 4)}, \qquad (\nu > 2), \tag{S.56}$$

Evaluating Eq. (S.55) at $\omega^\star(k)$ gives the minimal value of the critical line,

$$\lambda_c^\star(k) = \frac{1}{k}\,\lambda_c^\star(1) = \frac{\mu}{k}\,\frac{2^{\nu/2} - 1}{2^{\nu-1}}. \tag{S.57}$$

Note that by setting $k = 1$ in Eqs. (S.55), (S.56) and (S.57) we recover Eqs. (3) and (4) of the main text.

SUPPLEMENTARY NOTE 4: EFFECT OF MEMBERSHIP AND GROUP SIZE ON THE CRITICAL BEHAVIOUR

Effect of membership

In the main text case $g(k) = \delta_{k,1}$, i.e., individuals participate in only one group at a time. However, modern social settings often involve digital communication, allowing individuals to receive multiple stimuli in parallel. In our model, this corresponds to $\langle k \rangle > 1$, and Supplementary Fig. 1 illustrates how this reshapes the phase portrait for fixed group size $p(n) = \delta_{n,3}$ and $p(n) = \delta_{n,5}$.

For nonlinear contagion, increasing k progressively alters the influence of temporality. When $g(k) = \delta_{k,1}$ (see Fig. 2 and Supplementary Fig. 1a), the invasion threshold $\lambda_c(\omega)$ exhibits the non-monotonic behavior: it decreases with ω, reaches a finite minimum, and then increases again. When some individuals can participate in two groups (Supplementary Fig. 1b), this minimum shifts toward $\omega = 0$. For $g(k) = \delta_{k,3}$ (Supplementary Fig. 1c), the minimum disappears for sufficiently large nonlinearities, and the threshold becomes strictly increasing in ω: temporal reshuffling now consistently hinders adoption by disrupting the stability needed for reinforcement to accumulate. The same behavior is observed in Supplementary Fig. 1d-f for $p(n) = \delta_{n,5}$.

This qualitative change is can be captured analytically by the critical membership, since the case where the non–monotonicity disappears corresponds to $\omega^\star(k_c) = 0$. Imposing this condition in Eq. (S.56) gives

$$2^{\nu/2+1} + 2^{\nu+1} - 4 - k_c(\nu)\,(2^\nu - 4) = 0,$$

so that

$$k_c(\nu) = \frac{2^{\nu/2+1} + 2^{\nu+1} - 4}{2^\nu - 4}, \tag{S.58}$$

and its shown in the inset of Supplementary Fig. 1a. For $k < k_c(\nu)$, the threshold given by Eq. (S.55) remains non-monotonic and displays an optimal turnover rate. For $k > k_c(\nu)$, $\lambda_c(\omega, k)$ becomes strictly increasing: temporality always suppresses the onset of contagion when starting from a susceptible population. Since $k_c(\nu)$ approaches a horizontal asymptote in $k = 2$, even modest parallel exposure is enough to reverse the role of plasticity when synergy is strong.

Effect of group size

The comparison between top ($p(n) = \delta_{n,3}$) and bottom ($p(n) = \delta_{n,5}$) rows of Supplementary Fig. 1 shows how group size shapes: (*i*) the onset of non–monotonicity, i.e., the appearance of a minimum in $\lambda_c(\omega)$; and (*ii*) the value of synergy exponent ν at which that minimum is pushed to negative plasticity rates, $\omega^\star < 0$, making the curve monotonic again for all physically allowed $\omega > 0$. The first marks the appearance of a turning point in the invasion curve, while the second erases it.

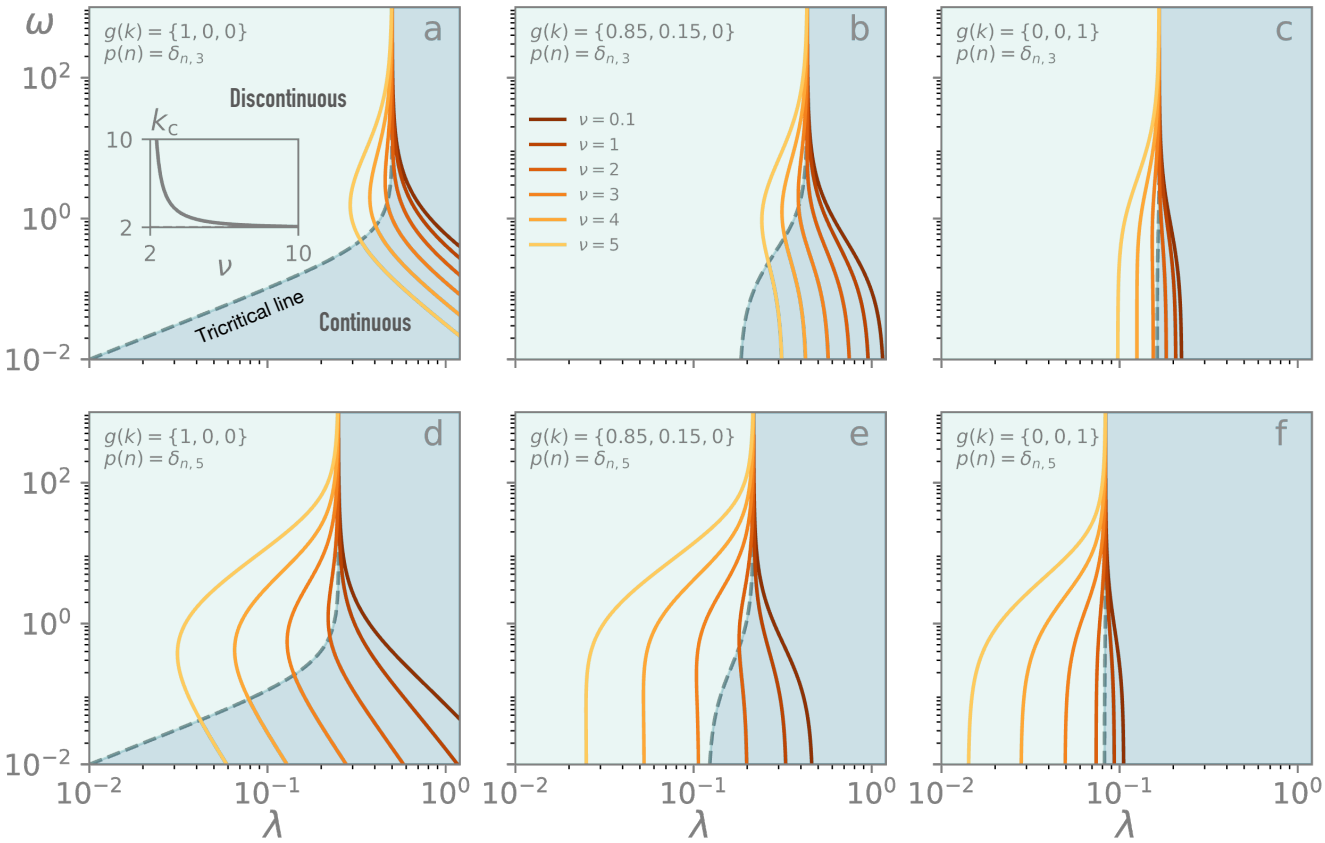
For $p(n) = \delta_{n,3}$ and low k, the invasion curve remains non–monotonic for all synergy values $\nu \geq 2$, with its minimum being at $\omega^\star > 0$. By contrast, for $p(n) = \delta_{n,5}$ the minimum appears for lower values of ν, and rapidly moves to $\omega^\star < 0$, so that $\lambda_c(\omega)$ becomes strictly decreasing for all $\omega > 0$. Therefore, n also reduces the relevance of temporality in determining the order of the transition.

To explain the effect of n in the monotonicity of the invasion threshold curve, we simplify Eq. (S.53) for a single membership ($g(k) = \delta_{k,1}$) and fixed group size $p(n) = \delta_{n,n_0}$. Thus, the invasion threshold satisfies

$$F_n(x) = \frac{\mu}{\omega}, \qquad x \equiv \frac{\lambda_c}{\mu + \omega}, \tag{S.59}$$

with

$$F_n(x) = n \sum_{i=1}^{n-1} \binom{n-1}{i} (i!)^\nu x^i. \tag{S.60}$$



Supplementary Fig. 1. **Interplay between membership, group size, plasticity, and synergy reshapes the phase diagram.** Phase portraits showing the critical adoption threshold λ_c as a function of the plasticity rate ω for different synergy factors, group sizes, and membership distributions. Panels **a–c** (**d–f**) correspond to fixed group size $p(n) = \delta_{n,3}$ ($p(n) = \delta_{n,5}$), while the left, central, and right columns correspond to $\langle k \rangle = 1.00$, 1.15, and 3.00, respectively. Each panel highlights the tricritical lines separating regions with continuous and discontinuous transitions. The inset in panel **a** shows the critical membership required to erase the non–monotonicity in the invasion threshold according to Eq. (S.58).

The invasion threshold becomes non–monotonic when $\lambda_c(\omega)$ develops a minimum. This occurs when

$$F_n'(x^\star) = 0, \qquad F_n(x^\star) = \frac{\mu}{\omega^\star}. \tag{S.61}$$

The condition $F_n'(x^\star) = 0$ selects the extremum, and the second relation determines its position in (λ, ω)–space. Because the combinatorial prefactors $n\binom{n-1}{i}(i!)^\nu$ grow rapidly with n for all $i \geq 2$, the higher-order terms of F_n and F_n' dominate much earlier for larger n. This produces two distinct consequences: On the one hand, there is an earlier onset of non-monotonicity: a positive solution $x^\star > 0$ to $F_n'(x) = 0$ appears for substantially smaller ν when the group size increases (explaining why the curves for $n = 5$ bend at lower synergy than for $n = 3$). On the other hand, the minimum of $\lambda_c(\omega)$ is shifted towards negative values of the plasticity rate. Once a positive extreme $x^\star$ exists, the corresponding $\omega^\star = \mu/F_n(x^\star)$ decreases sharply with n.

Explicitely, for $p(n) = \delta_{n,3}$, the polynomial

$$F_3(x) = 3\,[2x + 2^\nu x^2], \qquad F_3'(x) = 3\,[2 + 2^{\nu+1} x],$$

admits no positive stationary point, since $F_3'(x) = 0$ yields $x^\star = -2^{-\nu}$, which lies outside the physical domain. Consequently, $F_3(x^\star)$ never attains values large enough to drive $\omega^\star = \mu/F_3(x^\star)$ negative. The minimum of $\lambda_c(\omega)$ therefore always occurs at $\omega^\star > 0$, and the curve remains non–monotonic for all ν, consistent with the divergence of the required membership in Eq. (7).

In contrast, for $p(n) = \delta_{n,5}$,

$$F_5(x) = 5\left[\binom{4}{1}x + \binom{4}{2}2^{\nu}x^2 + \binom{4}{3}6^{\nu}x^3 + \binom{4}{4}24^{\nu}x^4\right],$$

where the cubic and quartic contributions grow rapidly with ν. As a consequence, $F_5'(x) = 0$ admits a positive solution $x^\star > 0$ already for moderate synergy. Moreover, the corresponding value $F_5(x^\star)$ increases sharply with ν, so that $\omega^\star = \mu/F_5(x^\star)$ can become negative. When this occurs, the minimum is pushed outside the physical domain $\omega > 0$, and the invasion threshold becomes strictly decreasing, exactly as observed in the bottom row of Supplementary Fig. 1.

SUPPLEMENTARY NOTE 5: PERSISTENCE THRESHOLD

We derive the analytical expression of the persistence threshold $\lambda_{\rm p}(\omega,\mu)$ for a hypergraph with three-body interactions ($p(n)=\delta_{n,3}$) under the single-membership condition ($g(k)=\delta_{k,1}$), keeping an explicit recovery rate $\mu>0$ throughout.

Stationary relations

Let f_i denote the stationary probability that a triplet contains $i=0,1,2,3$ infected individuals. The detailed-balance relations between consecutive occupancy states are

$$(i+1)\,\alpha\, f_{i+1} = (3-i)\,(\beta_i+\gamma)\, f_i, \qquad i=0,1,2, \tag{S.62}$$

where $\alpha=\mu+\omega(1-I)$, $\gamma=\omega I$. For single membership, the stationary node equation $s_1=\mu/(\mu+r)$ gives $I=1-s_1=\frac{r}{\mu+r}$. Inside each triplet the synergistic infection rates are

$$\beta_0=0, \qquad \beta_1=\lambda, \qquad \beta_2=2^{\nu}\lambda.$$

Iterating (S.62), all f_i can be written in terms of f_0:

$$f_1=\frac{3\gamma}{\alpha}f_0, \qquad f_2=\frac{\lambda+\gamma}{\alpha}f_1, \qquad f_3=\frac{2^{\nu}\lambda+\gamma}{3\alpha}f_2. \tag{S.63}$$

The effective infection pressure r reads

$$r=\mathcal{M}[I(r)]=\frac{\displaystyle\sum_{i=0}^{3}(3-i)\,\beta_i f_i}{\displaystyle\sum_{i=0}^{3}(3-i)\,f_i}. \tag{S.64}$$

Substituting (S.63) into (S.64), one finds that r satisfies the quadratic equation

$$a_2(\lambda,\omega,\mu)\,r^2+a_1(\lambda,\omega,\mu)\,r+a_0(\lambda,\omega,\mu)=0, \tag{S.65}$$

where, up to an overall common prefactor,

$$a_2(\lambda,\omega,\mu)=(\mu+\omega)^2+\lambda\,\omega, \tag{S.66}$$

$$a_1(\lambda,\omega,\mu)=-2^{\nu}\lambda^2\omega-2^{\nu}\lambda\,\omega^2-\lambda\,\mu\,\omega+2\mu(\mu+\omega)^2, \tag{S.67}$$

$$a_0(\lambda,\omega,\mu)=-\Big(2^{\nu}\lambda^2\mu\,\omega+2\lambda\mu^2\omega+2\lambda\mu\omega^2-\mu^2(\mu+\omega)^2\Big). \tag{S.68}$$

Equation (S.65) describes the non-zero stationary states of the dynamics; the absorbing branch $r=0$ has been factored out.

Closed expression of the persistence threshold

For fixed (ω,μ), the boundary of bistability corresponds to a saddle-node bifurcation of (S.65), i.e. to

$$\Delta_r(\lambda,\omega,\mu)=a_1(\lambda,\omega,\mu)^2-4\,a_2(\lambda,\omega,\mu)\,a_0(\lambda,\omega,\mu)=0. \tag{S.69}$$

Expanding (S.69) in powers of λ yields a quadratic equation of the form

$$A_{\rm p}(\omega,\mu)\,\lambda^2+B_{\rm p}(\omega,\mu)\,\lambda+C_{\rm p}(\omega,\mu)=0, \tag{S.70}$$

with coefficients

$$A_{\mathrm{p}}(\omega,\mu) = 2^{\nu+1}\big(2^{\nu}\omega + 3\mu\big), \tag{S.71}$$

$$B_{\mathrm{p}}(\omega,\mu) = 4^{\nu}\omega^2 + 2^{\nu+1}\mu\,\omega + 8\mu\,\omega + 9\mu^2, \tag{S.72}$$

$$C_{\mathrm{p}}(\omega,\mu) = -4\big(2^{\nu}-2\big)(\omega+\mu)^2. \tag{S.73}$$

The persistence threshold is determined by the positive solution of (S.70):

$$\lambda_{\mathrm{p}}(\omega,\mu) = \frac{-B_{\mathrm{p}}(\omega,\mu) + \sqrt{B_{\mathrm{p}}(\omega,\mu)^2 - 4A_{\mathrm{p}}(\omega,\mu)C_{\mathrm{p}}(\omega,\mu)}}{2A_{\mathrm{p}}(\omega,\mu)}. \tag{S.74}$$

Note that the persistence threshold in Eq. (S.74) is obtained by projecting the stationary AME dynamics onto the effective infection pressure r and enforcing the saddle–node condition on the resulting quadratic equation. While this reduction becomes asymptotically exact in the annealed limit, it deviates from the full AME spinodal at finite ω. The exact bistability boundary over the entire parameter range can instead be determined directly from the stationary self-consistency condition in Eq. (14) of the main text, by imposing the saddle–node condition on the full stationary AME system.

Annealed limit

The asymptotic behaviour of the persistence threshold for $\omega \to \infty$ follows directly from the discriminant condition (S.69), using the explicit coefficients $A_{\mathrm{p}}(\omega,\mu)$, $B_{\mathrm{p}}(\omega,\mu)$ and $C_{\mathrm{p}}(\omega,\mu)$ defined above. Since these coefficients are at most quadratic in ω, the discriminant $\Delta_r(\lambda,\omega,\mu) = a_1^2 - 4a_2a_0$ is a quartic polynomial in ω. Expanding (S.69) for large ω and retaining the leading contribution, we obtain

$$\Delta_r(\lambda,\omega,\mu) = \omega^4\,\lambda\Big(4^{\nu}\lambda + 8\mu - 2^{\nu+2}\mu\Big) + \mathcal{O}(\omega^3).$$

Imposing $\Delta_r = 0$ in the highly–annealed limit and discarding the trivial solution $\lambda = 0$, we obtain

$$\lim_{\omega\to\infty} \lambda_p(\omega,\mu) = \mu\,\frac{4(2^{\nu}-2)}{4^{\nu}} = \mu\,\frac{4(2^{\nu}-2)}{2^{2\nu}}, \tag{S.75}$$

and therefore, for large ω the persistence threshold approaches a horizontal asymptote, as shown in Fig. 2d of the main text.

Annealed limit

The asymptotic behaviour of the persistence threshold when $\omega \to \infty$ can be obtained directly from the discriminant condition $\Delta_r(\lambda,\omega,\mu) = 0$. Since a_2, a_1 and a_0 are at most quadratic in ω, the discriminant

$$\Delta_r(\lambda,\omega,\mu) = a_1(\lambda,\omega,\mu)^2 - 4\,a_2(\lambda,\omega,\mu)\,a_0(\lambda,\omega,\mu)$$

is a quartic polynomial in ω. Expanding in powers of ω and keeping the leading term, one finds

$$\Delta_r(\lambda,\omega,\mu) = \omega^4\,\lambda\Big(4^{\nu}\lambda + 8\mu - 2^{\nu+2}\mu\Big) + \mathcal{O}(\omega^3), \tag{S.76}$$

where we used $4^{\nu} = 2^{2\nu}$. For the discriminant to vanish in the highly–annealed limit, the coefficient of ω^4 must be zero, which (discarding the trivial solution $\lambda = 0$) yields

$$4^{\nu}\lambda + 8\mu - 2^{\nu+2}\mu = 0. \tag{S.77}$$

Solving for λ we obtain

$$\lim_{\omega\to\infty} \lambda_p(\omega,\mu) = \mu\,\frac{4\big(2^{\nu}-2\big)}{4^{\nu}} = \mu\,\frac{4\big(2^{\nu}-2\big)}{2^{2\nu}}. \tag{S.78}$$

Therefore, the persistence line approaches, for large ω, the horizontal asymptote, as shown in Fig. 2d of the main text.

Width of the bistable region

We can obtain a closed expression for the width of the bistability area in the mean field limit by subtracting Eqs. (S.51) and (S.78). It reads

$$\lim_{\omega\to\infty} \Delta\lambda(\omega) = \mu\left(\frac{1}{2} - \frac{4(2^{\nu}-2)}{2^{2\nu}}\right) = \frac{\mu}{2}\left(1 - 2^{\,2-\nu}\right)^2 .$$

SUPPLEMENTARY NOTE 6: REAL-WORLD STRUCTURES

This Supplementary Note details the analysis of seven face-to-face proximity datasets collected in different social environments: a primary school [1], a conference [2], a hospital [3], a village in Malawi [4], two editions of a workplace study [5], and a high school [6] (see Table I). All datasets have finite temporal resolution (20 s), and all contacts recorded within the same window are considered simultaneous. Below, we describe the preprocessing pipeline, the extraction of instantaneous group structures, the tracking of groups across time, and the estimation of the effective plasticity rate and the empirical distributions $\{p(n)\}$ and $\{g(k)\}$. These quantities form the structural input used in Fig. 3 and Fig. 4 of the main text.

Dataset	Context	$\langle \tau \rangle$ (s)	$\langle \omega \rangle$ (s^{-1})	Reference
InVS15	Workplace	1.89×10^2	5.3×10^{-3}	[5]
LyonSchool	Primary school	1.05×10^2	9.5×10^{-3}	[1]
LH10	Hospital	1.41×10^2	7.1×10^{-3}	[3]
Thiers13	High school	2.08×10^2	4.8×10^{-3}	[6]
SFHH	Scientific conference	1.30×10^2	7.7×10^{-3}	[2]
Malawi	Village	1.61×10^2	6.2×10^{-3}	[4]

Supplementary Table I. **Real-world face-to-face interaction datasets**. For each dataset we report its social context, the average residence time $\langle \tau \rangle$, the effective plasticity rate $\langle \omega \rangle$ measured from the data, and the corresponding reference.

Preprocessing of real face-to-face proximity data

Following the protocol of Iacopini et al. [7], we preprocess the raw interaction record to ensure temporal consistency and remove unreliable detections.

Data Cleaning: For each dataset, we (*i*) remove invalid interactions where an individual scans itself or the identifiers are missing; (*ii*) perform gap filling so that if an interaction (i, j) is observed at times $t - 1$ and $t + 1$ but missing at t, we impute a contact at t; (*iii*) discard transient contacts appearing at a single isolated timestamp; and (*iv*) apply triadic closure; adding a link (i, k) whenever at time t we observe (i, j) and (j, k), to merge fragmented detections.

Construction of Instantaneous Graphs: For each discrete timestamp t, we construct an undirected graph $G_t = (V_t, E_t)$ where V_t is the set of active individuals during window t, and $E_t = \{(i, j)\}$ contains all interactions recorded within that time window.

Group Extraction: Groups at time t are defined as maximal cliques of G_t, retaining only cliques of size $n \geq 2$. Each time slice is described by a collection of sets $\mathcal{G}_t = \{g_t^{(1)}, g_t^{(2)}, \dots\}$, where each $g_t^{(k)}$ represents a simultaneously interacting group.

Tracking groups in real-world structures

Since the groups at each timestamp are anonymous, we assign persistent identifiers to track their evolution across time.

Group tracking across time: Let $g_t \equiv g_t^{(k)}$ and $g_{t-1} \equiv g_{t-1}^{(l)}$ denote two group realizations at consecutive timestamps.Group identity is preserved whenever the two groups share their full core membership. Formally, we quantify this through the overlap similarity:

$$\mathrm{sim}(g_t, g_{t-1}) = \frac{|g_t \cap g_{t-1}|}{\min(|g_t|, |g_{t-1}|)}. \tag{S.79}$$

By construction, $\mathrm{sim} = 1$ whenever one group is a superset of the other. We link the pair (g_t, g_{t-1}) if

$$\mathrm{sim}(g_t, g_{t-1}) \geq \theta, \qquad |g_t \cap g_{t-1}| \geq 2,$$

using $\theta = 1$ in this work. Thus, group identity is preserved even if peripheral members join or leave, provided that the smallest group of at least two individuals is fully contained in the largest one.

All admissible matches with $\mathrm{sim} \geq \theta$ and $|g_t \cap g_{t-1}| \geq 2$ are computed, and non-conflicting assignments are selected via a greedy maximum-similarity rule. Groups at time t with no match receive new identifiers.

Residence times: For each individual i, we record the intervals of consecutive timestamps during which it remains in a group with the same persistent identity. The residence time is defined as

$$\tau_e = t_{\mathrm{end}} - t_{\mathrm{start}},$$

and the multiset $\{\tau_e\}$ is computed across all nodes and all such intervals. Because reporting occurs at 20s intervals, residence times are converted to seconds by multiplying by $\Delta t = 20$s.

Effective plasticity rate

The effective plasticity rate $\langle\omega\rangle$ quantifies the typical frequency with which individuals reshuffle between groups. Following the definition used in the main text (Eq. (8)), we compute

$$\langle\omega\rangle = \frac{1}{\langle\tau\rangle}, \qquad \text{being } \langle\tau\rangle = \frac{1}{E}\sum_{e=1}^{E}\tau_e, \tag{S.80}$$

where E is the total number of observed residence time intervals. Supplementary Table I reports measured values of $\langle\tau\rangle$ and $\langle\omega\rangle$, which vary widely across contexts, spanning almost an order of magnitude.

Size and membership distributions

Given the temporal sequence of group configurations $\{\mathcal{G}_t\}$, we extract the time series of group counts and the empirical distributions $p(n)$ and $g(k)$.

Time series of group counts: The raw timestamps may contain gaps. Therefore, we first define the full axis $[t_{\min}, t_{\max}]$ and assign an empty group list $\mathcal{G}_t = \emptyset$ to missing snapshots. Then, for each snapshot t, we compute the number of groups

$$N(t) = |\mathcal{G}_t|,$$

which is used for visualization (first column of Supplementary Fig. 2).

Group-size distribution $p(n)$: For all groups across all snapshots, we collect the set of sizes

$$\mathcal{S} = \{\, |g_t| : g_t \in \mathcal{G}_t \}.$$

If c_n denotes the number of occurrences of size n in $\mathcal{S}$, the normalized group-size distribution is

$$p(n) = \frac{c_n}{\sum_m c_m}.$$

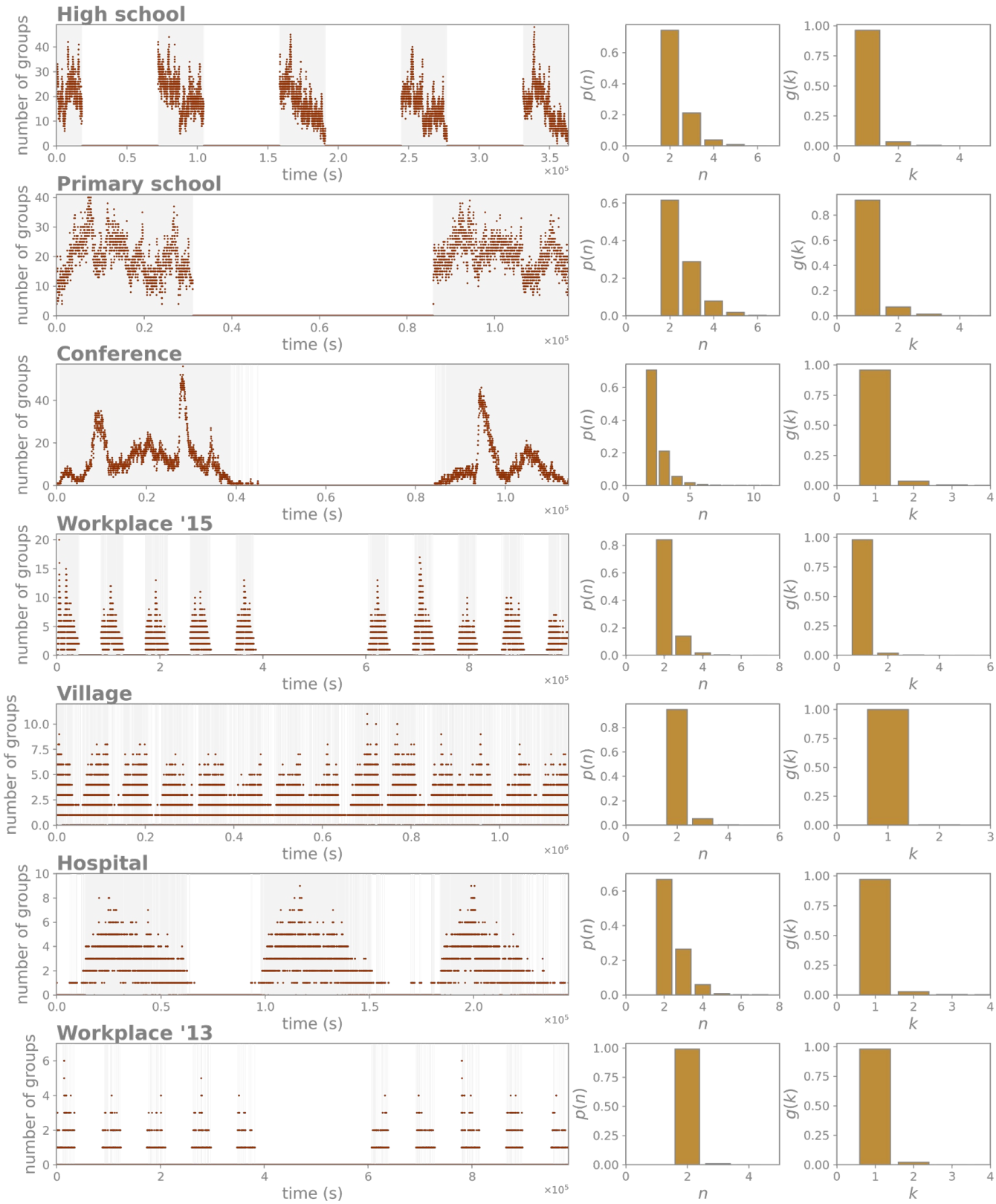
Membership-per-snapshot distribution $g(k)$: For each individual i at time t, we compute the number of groups the individual participates in,

$$k_i(t) = \big|\{\, g_t \in \mathcal{G}_t : i \in g_t \,\}\big|.$$

We collect the values $k_i(t)$ across all snapshots into the multiset $\mathcal{K}$, and let d_k denote the total number of occurrences of value k in $\mathcal{K}$. Then, the normalized distribution of memberships per snapshot is

$$g(k) = \frac{d_k}{\sum_m d_m}.$$

Both normalized distributions are shown for every dataset in Supplementary Fig. 2 and feed the expressions of the tricritical point and invasion threshold to produce Fig. 3 and Fig. 4.



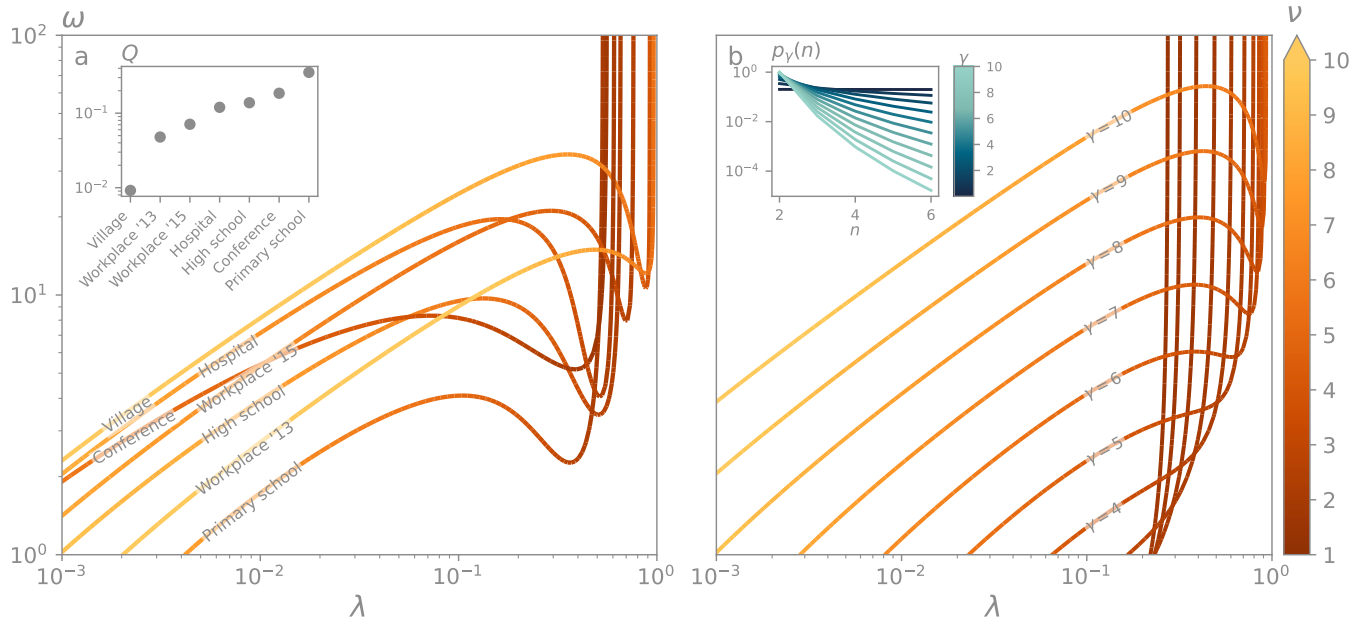
Supplementary Fig. 2. **Empirical temporal structure of real datasets.** Time series of the number of groups $N(t)$, distribution of group sizes $p(n)$, and distribution of memberships per snapshot $g(k)$. The datasets belong to the different social contexts reported in Table I.

SUPPLEMENTARY NOTE 7: EFFECT OF TEMPORALITY AND HETEROGENEITY IN REAL-WORLD AND SYNTHETIC STRUCTURES

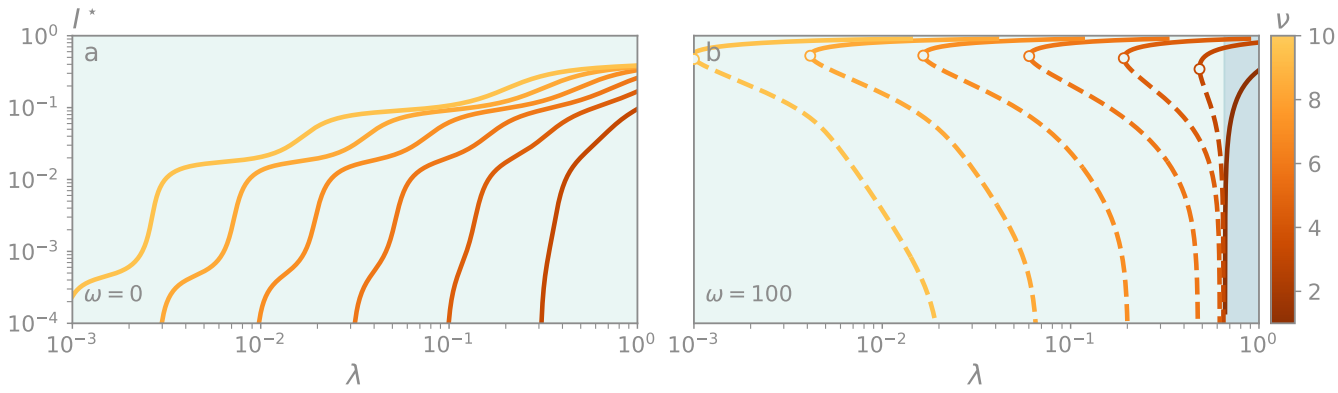
In Supplementary Fig. 3a we extend Fig. 3a of the main text by representing the tricritical lines for all the empirical face-to-face interaction datasets considered in Table I. We show that the qualitative phenomenology reported in Fig. 3a is robust across structures with different levels of size heterogeneity and plasticity. While the precise location of the tricritical line shifts from dataset to dataset, its shape remains unchanged. Furthermore, the inset reports the actual numerical values of the effective structural coupling Q (beyond the rank presented in Fig. 4).

To further disentangle the role of heterogeneity in the onset of multistability, in Supplementary Fig. 3b we perform a systematic analysis on synthetic structures with varying group size heterogeneity. We fix the membership distribution to match that of the high-school dataset [6] and vary only the group-size distribution, imposing truncated power-law forms $p(n) \sim n^{-\gamma}$ with cutoff $n_{\text{max}} = 6$ (consistent again to the high-school dataset, see Supplementary Fig. 2). Tuning the level of heterogeneity through the exponent γ, we observe that increasing heterogeneity turns the tricritical line from being a monotonous function to a non-monotonous one.

Finally, Supplementary Fig. 4 complements Fig. 3b by showing the quenched ($\omega = 0$ in Supplementary Fig. 4a) and close-to-annealed ($\omega = 100$ in Supplementary Fig. 4b) phase diagrams. The quenched limit displays continuous transitions with non-monotonous prevalence growth, showcasing different plateaus corresponding to different levels of mesoscale localization. This is consistent with previous findings by St-Onge et al. [8, 9] on the role of heterogeneity on the phase transition. The annealed limit displays discontinuous transitions, equivalent to the phenomenology obtained by Iacopini et al. [10]. This highlights that multistable active phases with multiple coexisting active states are observed in neither the annealed nor the quenched limits.



Supplementary Fig. 3. **Interplay between temporality and size heterogeneity yields multistability.** **a** Tricritical lines (solid curves) in the (λ, ω) plane for the empirical face-to-face datasets in Table I. The inset reports the effective structural coupling of each dataset according to Eq. (25). **b** Tricritical lines for synthetic structures with fixed membership distribution (matching the high-school dataset) and power-law group-size distributions $p(n) \sim n^{-\gamma}$ with cutoff $n_{\max} = 6$ (reported in the inset).



Supplementary Fig. 4. **Cross-sections of the phase diagram at fixed plasticity rates.** Stationary prevalence $I^\star$ as a function of the adoption rate λ for increasing values of the synergy exponent ν (color-coded) in the quneched ($\omega = 0$, **a**) and close-to-annealed ($\omega = 100$, **b**) limits. Solid (dashed) lines denote stable (unstable) stationary states.

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SUPPLEMENTARY NOTE 8: EFFECT OF HOMOGENIZING TEMPORALITY IN CHOWDHARY ET AL.

In Ref. [11], Chowdhary et al. analyse simplicial contagion on higher-order networks, and their Fig. 1 shows that temporality hinders the onset of contagion dynamics, even in the linear case. In Supplementary Fig. 5 we show that this effect does not originate from temporality itself, but from two exogenous factors introduced by their modelling framework.

First, their temporal model rewires the structure at every time step without preserving the degree sequence. As a result, node degrees fluctuate across time and across nodes. This induces artificial structural heterogeneity that does not stem from temporality, but from repeatedly resampling the network from scratch.

Second, they use a Markov chain approach (MMCA) that assumes independent contagion pathways. Specifically, the probability that one node i out of the N nodes in a structure is infected at time $t+1$ is assumed to follow

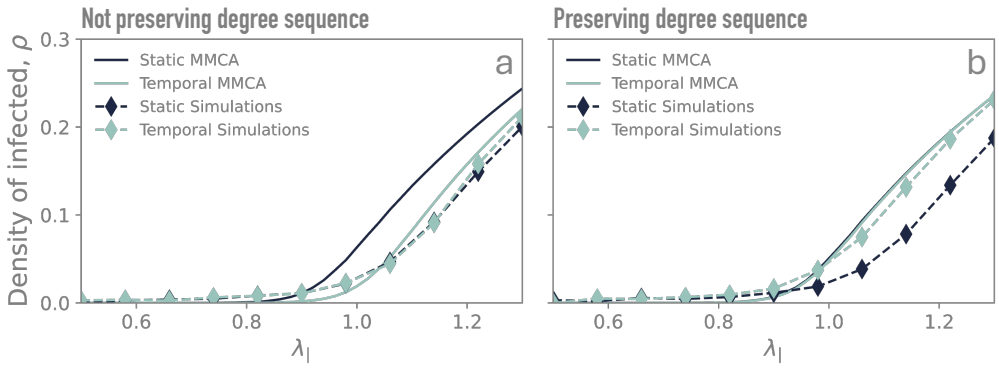
$$p_i(t+1) = \big(1 - q_i(t)\, q_{i,\Delta}(t)\big)\big(1 - p_i(t)\big) + (1-\mu)\, p_i(t), \tag{S.81}$$

where $p_i(t)$ is the infection probability of node i at time t, μ is the recovery probability, $q_i(t)$ is the probability that i is not infected via pairwise interactions with its neighbours, and $q_{i,\Delta}(t)$ is the probability that i is not infected via any of its two-simplices. These are approximated as

$$q_i(t) = \prod_{j \in \Gamma_i(t)} \big(1 - \beta\, p_j(t)\big), \tag{S.82}$$

$$q_{i,\Delta}(t) = \prod_{j,\ell \in \Delta_i(t)} \big(1 - \beta_\Delta\, p_j(t)\, p_\ell(t)\big), \tag{S.83}$$

where $\Gamma_i(t)$ is the set of edges attached to i, at time t, and $\Delta_i(t)$ the set of triangles incident to i. The constants β and β_Δ are the infection probabilities associated with edges and triangles, respectively.



Supplementary Fig. 5. **Effect of preserving the degree sequence on temporal SIS dynamics.** leyenda de arriba al reves Panels **a** and **b** compare the stationary prevalence $\rho = N^{-1}\sum_i p_i$ as a function of the infection rate $\lambda = \beta\langle k\rangle/\mu$ for static and temporal networks in the linear case ($\beta_\Delta = 0$). Temporal networks are generated by randomly rewiring the edges at each step. In **a**, the rewiring does not preserve the degree sequence, so node degrees fluctuate in time, as in the temporal model of Ref. [11]. Under these conditions, the iteration of the Markov chain equations (S.81)–(S.83) predicts a suppression of contagion in the temporal setting [11]. In **b**, the rewiring does preserve the degree sequence at each snapshot. In this case, the Markov chain approach yields identical prevalence curves for the static and temporal networks, demonstrating that the suppression observed in **a** arises from temporal degree fluctuations rather than temporality itself. Monte Carlo simulations (markers) show the opposite trend: when the degree sequence is preserved, temporality shifts the threshold to lower values of λ, in agreement with known results on temporal networks [12, 13]. When the degree sequence is not preserved (panel **a**), this structural heterogeneity partly compensates this effect, but the temporal curve remains above the static one in the supercritical regime. All structures are Random Simplicial Complexes (RSC, see Ref. [10]), have $N = 500$ nodes and average degree $\langle k\rangle = 12$. We consider the linear case $\beta_\Delta = 0$, and the recovery rate is set to $\mu = 0.1$, and the markers of the Monte Carlo simulations represent the average of 1000 realizations.

Supplementary Fig. 5a, compares the stationary state $\rho = N^{-1}\sum_i p_i$ obtained from the iteration of Eqs. (S.81)-(S.82) on a static structure with the corresponding temporal dynamics when the degree sequence is not preserved.

In the linear case ($\beta_\Delta = 0$), the result reproduces the qualitative trend reported in Ref. [11]: the temporal structure appears to hinder contagion.

However, Supplementary Fig. 5b shows the same comparison when the temporal evolution preserves the degree sequence. In this case, the Markov chain approximation predicts identical stationary states for the static and temporal networks. This demonstrates that the differences observed in Supplementary Fig. 5a are fully driven by the temporal fluctuations in the degrees of the nodes, and not by temporality itself.

Finally, Supplementary Fig. 5 also includes stochastic Monte Carlo simulations. When the degree sequence is preserved (Supplementary Fig. 5b), temporality shifts the epidemic threshold to lower values of λ, in agreement with the well-established behaviour of temporal networks [12, 13]. When the degree sequence is not preserved (Supplementary Fig. 5a), the imposed degree fluctuations partially counteract the facilitation effect of temporality, but the temporal prevalence curve still lies above the static one in the supercritical regime.

SUPPLEMENTARY REFERENCES

[1] J. Stehlé, N. Voirin, A. Barrat, C. Cattuto, L. Isella, J.-F. Pinton, M. Quaggiotto, W. Van den Broeck, C. Régis, B. Lina, et al., PloS ONE **6**, e23176 (2011).

[2] L. Isella, J. Stehlé, A. Barrat, C. Cattuto, J.-F. Pinton, and W. Van den Broeck, Journal of Theoretical Biology **271**, 166 (2011).

[3] P. Vanhems, A. Barrat, C. Cattuto, J.-F. Pinton, N. Khanafer, C. Régis, B.-a. Kim, B. Comte, and N. Voirin, PloS ONE **8**, e73970 (2013).

[4] L. Ozella, D. Paolotti, G. Lichand, J. P. Rodríguez, S. Haenni, J. Phuka, O. B. Leal-Neto, and C. Cattuto, EPJ Data Science **10**, 46 (2021).

[5] M. Génois and A. Barrat, EPJ Data Science **7**, 1 (2018).

[6] R. Mastrandrea, J. Fournet, and A. Barrat, PloS one **10**, e0136497 (2015).

[7] I. Iacopini, M. Karsai, and A. Barrat, Nature Communications **15**, 7391 (2024).

[8] G. St-Onge, V. Thibeault, A. Allard, L. J. Dubé, and L. Hébert-Dufresne, Physical Review E **103**, 032301 (2021).

[9] G. St-Onge, I. Iacopini, V. Latora, A. Barrat, G. Petri, A. Allard, and L. Hébert-Dufresne, Communications Physics **5**, 25 (2022).

[10] I. Iacopini, G. Petri, A. Barrat, and V. Latora, Nature Communications **10**, 2485 (2019).

[11] S. Chowdhary, A. Kumar, G. Cencetti, I. Iacopini, and F. Battiston, Journal of Physics: Complexity **2**, 035019 (2021).

[12] E. Valdano, L. Ferreri, C. Poletto, and V. Colizza, Physical Review X **5**, 021005 (2015).

[13] G. St-Onge, J.-G. Young, E. Laurence, C. Murphy, and L. J. Dubé, Physical Review E **97**, 022305 (2018).